

RITHIK AGRAWAL

Senior Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer with 5 years of experience architecting, developing, and scaling high-performance full-stack applications in fintech and telecom domains. Deep expertise in Python (Flask, FastAPI), PostgreSQL, and Angular with microservices architecture and CI/CD automation. Proven track record of reducing API response times by 35%, improving deployment efficiency by 50%, and engineering platforms supporting millions of active users.

TECHNICAL SKILLS

Languages & Frameworks: Python, Flask, FastAPI, Angular, TypeScript, JavaScript, RxJS, HTML5/CSS3, Bootstrap, Angular Material
Databases & Caching: PostgreSQL, SQL, Redis, Data Modeling, Query Optimization, Alembic
Architecture & APIs: Microservices, RESTful APIs, JWT, OAuth 2.0, RBAC, High-Level & Low-Level Design (HLD/LLD), Swagger/OpenAPI
DevOps & Cloud: AWS (EC2, RDS, S3), DigitalOcean, Docker, Jenkins, GitHub Actions, Linux, Nginx, CI/CD Automation
Testing & Methodologies: PyTest, Unit & Integration Testing, Test-Driven Development (TDD), Agile/Scrum, Git, JIRA, Postman

PROFESSIONAL EXPERIENCE

Senior Software Developer

08/2022 – Present

Power Financial Wellness Inc

Pune, India

- Architected and deployed distributed backend microservices in **Python (Flask/FastAPI)**, orchestrating asynchronous event workflows and achieving **99.99% service availability** for 50,000+ monthly active users.
- Engineered 25+ mission-critical RESTful APIs featuring JWT authentication, granular Role-Based Access Control (RBAC), Redis-backed token-bucket rate limiting, and Pydantic validation schemas, mitigating unauthorized access by **100%**.
- Spearheaded end-to-end PostgreSQL schema architecture, designing B-Tree composite indexing, partial indexes, and partitioning strategies that slashed P95 database query latency by **35%** across high-volume transaction tables.
- Authored comprehensive High-Level Design (HLD) and Low-Level Design (LLD) architectural specifications for multi-tenant SaaS modules, establishing blueprints adopted across 3 cross-functional engineering pods.
- Constructed modular, reactive UI interfaces in **Angular 15+**, **TypeScript**, and **RxJS** state management, elevating component reuse across platforms by **40%** and reducing frontend defect rates by **30%**.
- Integrated asynchronous payment gateways, webhook ingestion pipelines, and CRM messaging platforms, processing recurring transactions with zero data loss and automated idempotency verification.
- Automated multi-stage CI/CD deployment pipelines utilizing Docker containerization, **Jenkins**, and automated blue-green releases, reducing build-to-deployment turnaround time by **50%** and eliminating release downtime.
- Instituted rigorous Test-Driven Development (TDD) protocols with **PyTest** unit and integration suites, elevating mission-critical backend code coverage from 55% to **85%+**.
- Mentored 3 associate developers through structured architectural code reviews, design critiques, and engineering best practices, accelerating sprint velocity by **25%**.

Software Engineer

08/2021 – 08/2022

Jio Platforms Limited

Mumbai, India

- Engineered resilient backend services and interactive frontend interfaces for **JioMeet**, India's premier video collaboration platform serving **15M+ active users**, managing real-time meeting session states and user metadata.
- Designed and deployed centralized OAuth 2.0 identity federation and single sign-on (SSO) workflows, safeguarding cross-service session validation for millions of concurrent connections.
- Automated database schema evolution and transactional data migrations using **Alembic** across PostgreSQL clusters, preventing schema drift and enabling zero-downtime database upgrades.
- Standardized reusable micro-frontend widgets and REST service wrappers, cutting code duplication by **30%** and accelerating sprint feature delivery across a 20-engineer squad.
- Enhanced system reliability and reduced production incident volume by **25%** by enforcing PyTest regression suites, mock testing harnesses, and structured daily peer reviews.

KEY PROJECTS

Service Booking and Management Platform | *Python, Flask, Angular, PostgreSQL, Redis, AWS, Docker* 08/2021 – Present

- Architected full-stack service booking system handling high-concurrency booking, scheduling, and payment processing.
- Designed multi-tier RBAC for Admin, Vendor, and Customer roles with granular access permissions and security audit logging.
- Implemented RESTful endpoints with cursor pagination, advanced filtering, and **Redis caching**, improving throughput by **40%**.
- Containerized application with Docker and deployed on **AWS (EC2, RDS, S3)** with automated CI/CD via GitHub Actions.

Admin and Vendor Management Portal | *Angular, TypeScript, REST APIs, Swagger, PostgreSQL*

- Engineered real-time operations dashboards featuring KPI reporting and interactive data visualization for enterprise stakeholders.
- Integrated automated email/SMS notifications, event audit logging, and external REST APIs to streamline operational workflows.
- Designed comprehensive Swagger/OpenAPI documentation for all endpoints, accelerating third-party partner integration.

EDUCATION

Rajiv Gandhi Proudhyogiki Vishwavidyalaya

Bachelor of Engineering in Computer Science

Bhopal, India

07/2017 – 06/2021